

Alireza Nasirzadeh, PhD

Research Professor



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EDUCATION

KHARAZMI UNIVERSITY

Doctor of Philosophy in Sports Biomechanics, 2016

Thesis title: Comparison of Selected Kinematical, Kinetic, and Functional Lower Limb Variables during Walking and Running at Different Speeds among Young Male Athletes Focusing on Limb Dominance (Mentor: [Prof Heydar Sadeghi](#)).

UNIVERSITY OF BIRJAND

Master of Science in Physical Education and Sports Sciences, 2012

Dissertation title: Architectural Characteristics of the Upper Extremity Skeletal Muscles in Young Male Table Tennis Players

UNIVERSITY OF BIRJAND

Bachelor of Science in Pure Mathematics, 2009

PROFESSIONAL EXPERIENCE

CHUNG-ANG UNIVERSITY

School of Mechanical Engineering,
Assistive and Rehabilitation Robotics Laboratory

2022 - current: [Research Professor](#)

I.R.I. NATIONAL PARALYMPIC COMMITTEE

Athlete Evaluation Department

2020 - 2021: Biomechanist

KHARAZMI UNIVERSITY

Faculty of physical education and sport sciences
Department of Sports Biomechanics

2019 - 2021: Assistant Professor

2016 - 2019: Teaching Assistant

PEER-REVIEWED PUBLICATIONS

Nasirzadeh, A. et al. (2026). [Biomechanical Effects of Passive Exosuit Assistance on Tibiofemoral Loading and Dynamic Stability during Downhill Walking](#). *Journal of NeuroEngineering and Rehabilitation*.

Park, C., **Nasirzadeh, A.**, Lee, K., Kim, G., & Lee, G. (2026). [Real-Time Estimation of User Adaptation During Hip Exosuit-Assisted Walking Using Wearable Inertial Measurement Unit Data and Long Short-Term Memory Modeling](#). *Biomimetics*, 11(2), 96.

Nasirzadeh, A., et al. (2025). [Harnessing Resistance: The Impact of a Passive Exosuit on Golf Swing Biomechanics](#). *Journal of Biomechanics*, 192, 112958.

Choe, E., Moon, J., Ryu, J., Yang, S., **Nasirzadeh, A.**, Kong, S., Choi, Y., & Lee, G. (2025). [Low-Profile, Shoe-Type Ankle-Foot Orthosis with Active Variable Ankle Stiffness via Wire-Fabric Compression Mechanism](#). *Biomimetics*, 10(8), 539.

Kim, Y.; Park, S.; Yang, S.; **Nasirzadeh, A.**; Lee, G. (2024). [The Interplay Between Muscular Activity and Pattern Recognition of Electro-Stimulated Haptic Cues During Normal Walking: A Pilot Study](#). *Bioengineering*, 11, 1248.

Nasirzadeh, A. et al. (2023). [The Effects of a Custom-Designed High-Collar Shoe on Muscular](#)

[Activity, Dynamic Stability, and Leg Stiffness: A Biomimetic Approach Study.](#) *Biomimetics*, 8, 274.

Park, J., Nam, K., Yun, J., Moon, J., Ryu, J., Park, S., **Nasirzadeh, A.** et al. (2023). [Effect of hip abduction assistance on metabolic cost and balance during human walking.](#) *Science Robotics*, 8(83), eade0876-eade0876.

Nasirzadeh, A. et al. (2023). [Influence of circumferential ankle pressure of shoe collar on the kinematics, dynamic stability, electromyography, and plantar pressure during normal walking.](#) *PLOS ONE*, 18(2), e0281684.

Nasirzadeh, A. (2020). [The influence of horizontal velocity on inter-lower-limbs local and global asymmetry during walking.](#) *Physical Treatments*, 10(2), 69-78.

Nasirzadeh, A., Mohseni, F. (2018). [Local and global symmetry during jogging in young male athletes.](#) *Physical Treatments*, 8(1), 17-26.

Nasirzadeh, A., et al. (2017). [Gait symmetry and its evaluation methods: a review.](#) *Scientific Journal of Rehabilitation Medicine*, 6(2), 283-97.

Nasirzadeh, A., et al. (2017). [A review of selected factors that affect gait symmetry.](#) *Physical Treatments*, 7(1), 3-12.

Nasirzadeh, A., et al. (2016). [The influence of movement velocity on inter-lower-limbs kinematical symmetry in normal walking: a pilot study.](#) *Scientific Journal of Rehabilitation Medicine*, 5(4), 159-72.

Nasirzadeh, A., et al. (2015). [Multivariate analysis of 200-m front crawl swimming performance in young male swimmers.](#) *Acta of Bioengineering and Biomechanics*, 17(3), 137-43.

Nasirzadeh, A., et al. (2014). [Selected anthropometrical, muscular architecture, and biomechanical variables as predictors of 50-m performance of front crawl swimming in young male swimmers.](#) *Science and Sports*, 29(5), e75-e81.

Nasirzadeh, A., et al. (2014). [Relationship between sprint performance of front crawl swimming and muscle fascicle length in young swimmers.](#) *Journal of Sports Science and Medicine*, 13(3), 550-556.

Under review or in progress

Kim, G., Lee, S., **Nasirzadeh, A.** (under review). Rapid Hip-Abduction Exosuit-Induced Torque to Improve Mediolateral Standing Balance Without Stepping. *Science advances*.

PRESENTATIONS

Nasirzadeh, A., et al. (2026). Mediolateral Balance Recovery during Walking using H-Medi Exosuit. 44th International Society of Biomechanics in Sports Conference, Loughborough, UK.

Nasirzadeh, A. (2025). Design and Validation of a Passive Exosuit to Enhance Golf Swing Performance. 7th International Conference on Medicine in Novel Technology and Devices, Hangzhou, China.

Nasirzadeh, A., et al. (2025). Reducing Tibiofemoral Joint Force During Downhill Walking Using Various Passive Exosuits: A Pilot Study. XXX Congress of the International Society of Biomechanics (ISB), Stockholm, Sweden.

Nasirzadeh, A., et al. (2024). [Power Up The Swing: A Biomechanical Study Towards A Golf Exosuit.](#) 42nd International Society of Biomechanics in Sports Conference, Salzburg, Austria.

Choe, E., Moon, J., Ryu, J., Yang, S., **Nasirzadeh, A.** (2024). [Active Variable Compression-Shoes: Design and Evaluation of an External Ankle Supporter to Prevent Ankle Sprain via Active Compression Mechanism.](#) 10th IEEE RAS/EMBS International Conference for Biomedical Robotics and Biomechatronics (BioRob), Heidelberg, Germany.

Nasirzadeh, A., et al. (2023). Passive Collar Stiffness of a Custom-designed High-Collar Shoe.

XXIX Congress of the International Society of Biomechanics (ISB), Fukuoka, Japan.

Nasirzadeh, A. (2021). Isokinetic analysis of elite Iranian Paralympic athletes. Second National Olympic & Paralympic Congress, Tehran, Iran.

Mohseni, F., **Nasirzadeh, A.** (2018). The effect of long-term custom-made foot orthoses on foot structures in children with flexible flat feet. Eleventh International Conference of Sport Sciences, Tehran, Iran.

Sobhkhiz, A., Farahpour, N., Sokhangooei, Y., **Nasirzadeh, A.** (2015). The effect of a researcher-made soft insole on selected kinematic and kinetic variables of walking of active girls. First National Conference on New Findings in Sport Sciences, Tehran, Iran.

Nasirzadeh, A., et al. (2013). The progress of front crawl swimming speed performance in adolescent swimmers emphasizes biomechanics, muscular architecture, and anthropometric factors. Second National Conference on Sports Talent Identification, Tehran, Iran.

Ilbeigi, S., Yousefi, M., **Nasirzadeh, A.**, Van Gheluwe, B. (2012). The effect of gender and anthropometrical parameters on the dynamic of sprint start among young elite sprinters. XXX International Society of Biomechanics in Sports (ISBS) Conference, Melbourne, Australia.

Ilbeigi, S., **Nasirzadeh, A.**, Ehsanbakhsh, A. (2012). Relationship between sprint performance of front crawl swimming and muscle fascicle length in young swimmers. XXXII FMSI World Congress of Sports Medicine, Rome, Italy.

Nasirzadeh, A., et al. (2011). Ultrasonographic examination of muscle architecture parameters of the upper arm muscles of table tennis players aged 40-60 years. Sixth National Conference on Physical Education and Sport Sciences, Tehran, Iran.

Nasirzadeh, A., et al. (2011). Investigating the indicators of muscular architecture in different parts of the brachial muscles of table tennis players. First International Sports Medicine Congress, Isfahan, Iran.

INVITED TALKS AND WORKSHOP EXPERIENCE

Epidemiology of Sports Injuries and Injury Prevention Strategies in Para Powerlifting (online), Asian Paralympic Committee, March 2021.

Physical Fitness Parameters in Children with Disabilities, I.R.I. Sports Federation for the Disabled, August 2018.

Principle of Motion Analysis and Interpreting Results using Visual 3D Software, Kharazmi University, February 2018.

Force Plate: Applications in Sport Sciences and Data Analysis, Shiraz University, October 2017.

Biomechanical Assessment of Athletic Performance, I.R.I. Sports Federation for the Disabled, January 2014.

TRANSLATION PUBLISHED IN THE PERSIAN LANGUAGE

Nasirzadeh, A. (2018). Walking Biomechanics. Sport Sciences Research Institute of Iran.

Nasirzadeh, A., Fattahi, A. (2015). Persian Translation of Vanlandewijck, Y. C., Thompson, D. W. R. (2011). Handbook of Sports Medicine and Science, The Paralympic Athlete. Sport Sciences Research Institute of Iran.

IlBeigi, S., **Nasirzadeh, A.**, Mohebbi, B., Araghanei, H. (2012). Persian Translation of Blazeovich, A. J. (2007). Sports Biomechanics: The Basics: Optimizing Human Performance. Sport Sciences Research Institute of Iran.

AWARDS AND HONORS

Kharazmi University
Outstanding Student, Department of Sports Biomechanics

2016

First National Conference on New Findings in Sport Sciences

Outstanding Presentation Award in Biomechanics 2015

Sixth National Conference on Physical Education and Sport Sciences
New Investigator Award 2011

PEER-REVIEWER

Journals

Journal of Sports Sciences, Sports Biomechanics, International Journal of Athletic Therapy & Training, Physical Treatments, PLOS ONE, IEEE Transactions on Neural Systems & Rehabilitation Engineering.

ENGLISH LANGUAGE SKILLS

PTE Academic - score out of 90
Listening: 85 Reading: 79 Writing: 80 Speaking: 90 Overall: 81 2024

GRE (General module) - score out of 170
Verbal Reasoning: 140 Quantitative Reasoning: 163 2021

IELTS (Academic module) - score out of 9
Listening: 8.5 Reading: 7.5 Writing: 6.5 Speaking: 7.0 Overall: 7.5
CEFR Level: C1 2021

LICENSES AND CERTIFICATES

Judo Coaching Certificate 2018
Judo Black Belt 1st Dan Certificate 2015
Swimming Pool Lifeguard Certificate 2011
Table Tennis Coaching Certificate 2009

TECHNICAL SKILLS

Programming languages: MATLAB, LabVIEW

Biomechanical software: Visual3D, AnyBody Modeling System™

Motion capture system: Vicon, Qualisys, OptiTrack

Force plate: AMTI, Kistler, AMTI Instrumented Treadmills

Electromyography: Delsys, Noraxon

IMU sensor: XSENSE

Foot scanner: Pedar, XSENSOR

Isokinetic: Biodex system 4

Force sensor: FUTEK

Data acquisition board: NATIONAL INSTRUMENTS NI myDAQ, USB-6255

Ultrasonography: Hitachi Aloka Medical

REFERENCES

Giuk Lee

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Department of Mechanical Engineering, Chung-Ang University
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Associate Professor,
College of Sport Sciences, Chung-Ang University
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Physical Education and Sports Sciences Department, University of Birjand
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